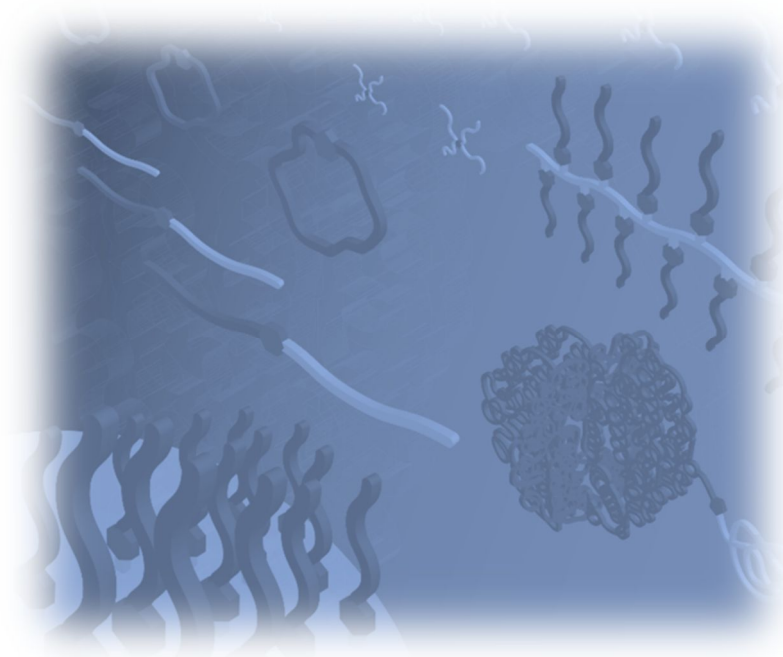


MoPoS – Makromolekulare Chemie II

Prof. Dr. Guillaume Delaître

Chair of Macromolecular Chemistry

Summer Semester 2026



The Module Part

Organization

Lecture *with me*

Every Thursday 10:30–12:00 in lecture hall 6G

except 14.05. (Ascension), 21.05. (absence), 04.06. (Corpus Christi), and 18.06. (absence)

→ In-person lectures of 21.05. and 18.06. will be replaced by video lectures.

Exercise (in German) *with Dr. Mina Tabatabai*

4 x 2 h (in principle Wednesdays at 8:30, first appointment May 13
– possible punctual swap between lecture and exercise)

Practical course (in German/English) *organized by PD Dr. Klaus Schaper*

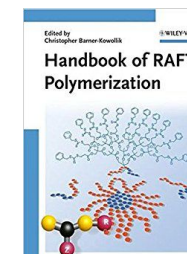
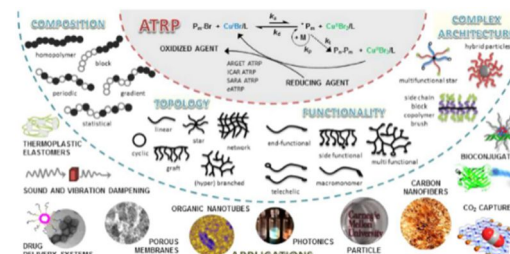
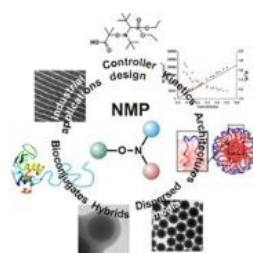
supervised, among others, by Dr. Seçkin Altuncu, Dr. Catarina Garcia Bravo, and Mr Gero Bramlage

Exam - July 30 and September 21

MSc Chemie 120 min (1/3, together with *Heterocyclenchemie* and *Stereochemie*)

MSc Wirtschaftschemie 120 Min (1/2, together with *Heterocyclenchemie*)

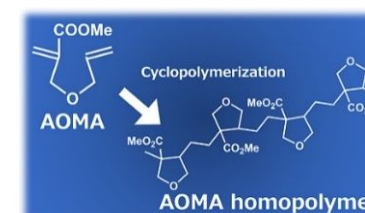
- 1] “Controlled radical polymerization”
Reversible-deactivation radical polymerization:
NMP, ATRP, and RAFT polymerisation



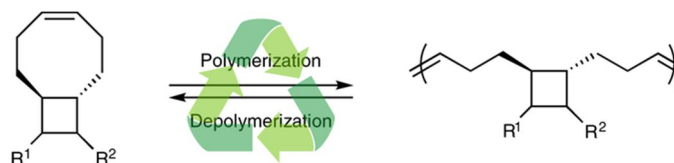
- 2] Insertion polymerization; synthesis of polyolefins



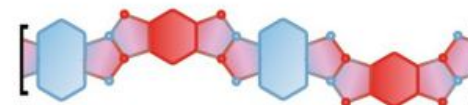
- 3] Cyclopolymerization and acyclic diene metathesis (ADMET) polymerization



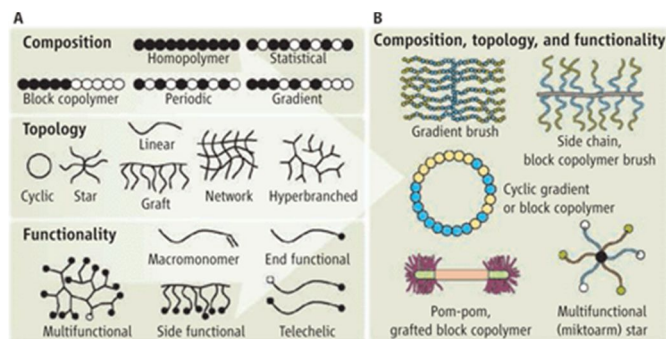
- 4] Ring-opening polymerizations



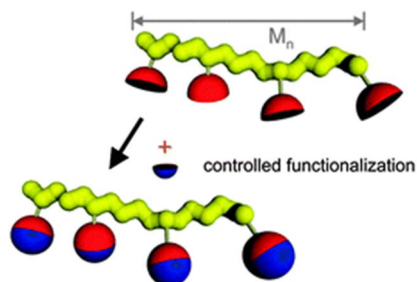
5] Conducting polymers and ladder polymers



6] Synthesis of comb and star polymers



7] Polymer analogous reactions



The Lecture

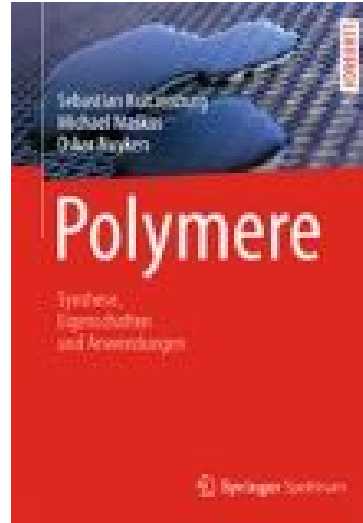
Recommended Literature

Polymere: Synthese, Eigenschaften und Anwendungen

von Sebastian Koltzenburg, Michael Maskos
und Oskar Nuyken

Springer

English version also available

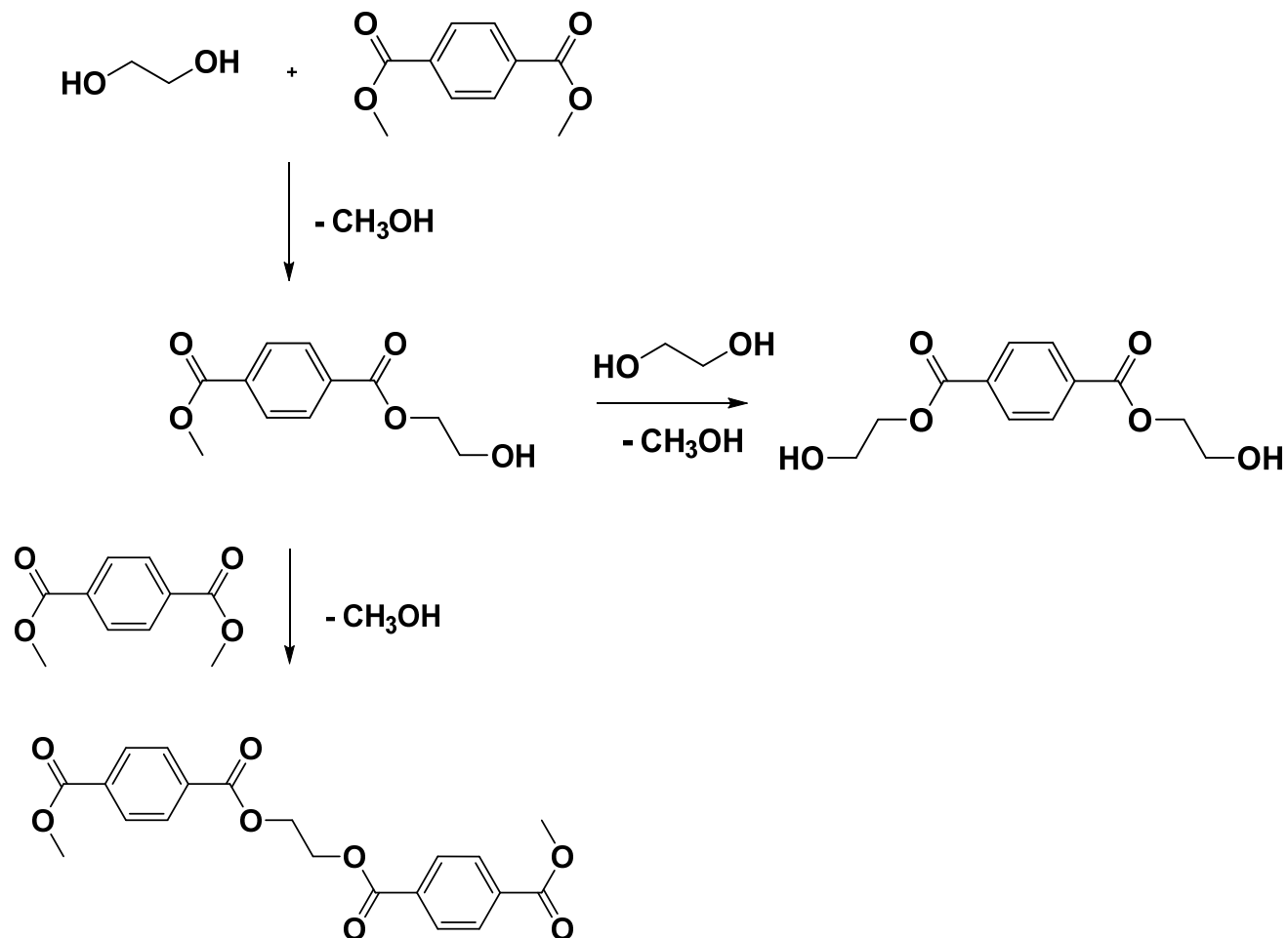


CHAPTER 0

Reminder

Polymerization Classes (in Short)

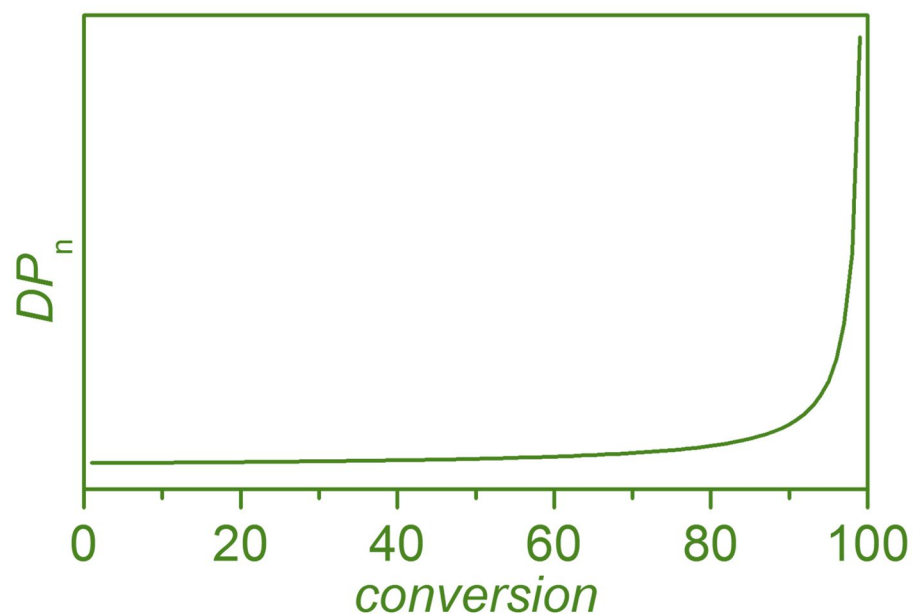
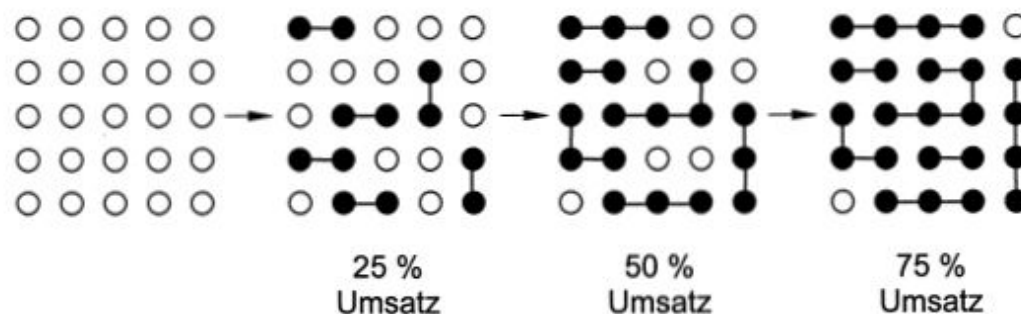
Step(-Growth) Polymerization



Polymerization Classes (in Short)

Step(-Growth) Polymerization

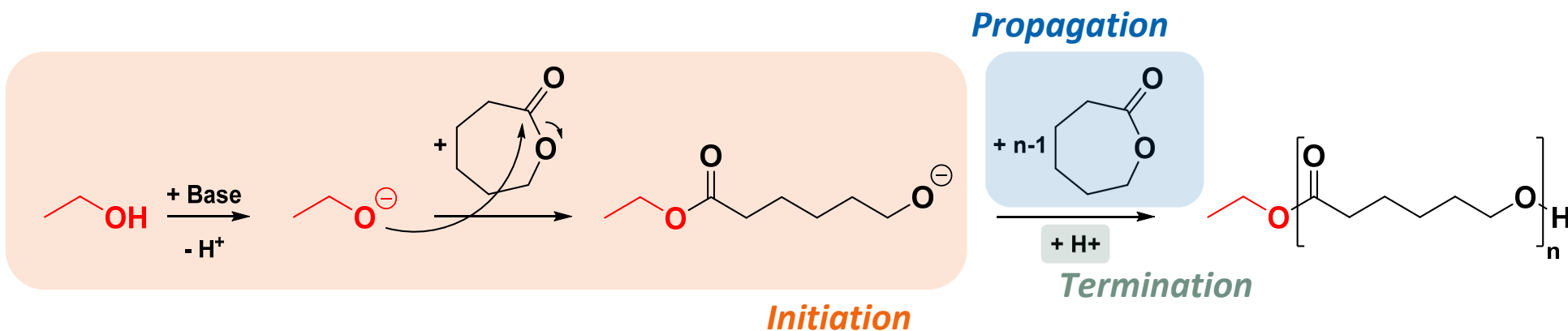
Step polymerizations are characterized by the disappearance of monomer very early in the reaction, far before the production of high-molecular-weight polymer.



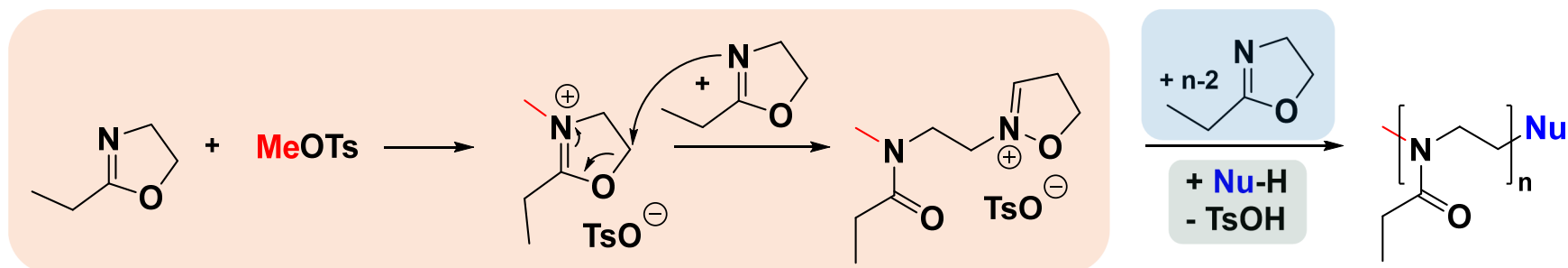
Polymerization Classes (in Short)

Chain(-Growth) Polymerization

Anionic Polymerization



Cationic Polymerization



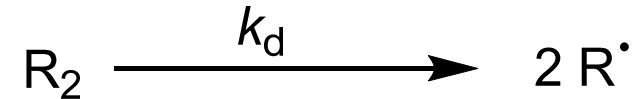
Ideally, for both mechanisms, the termination reaction occurs under human action (like here).

Polymerization Classes (in Short)

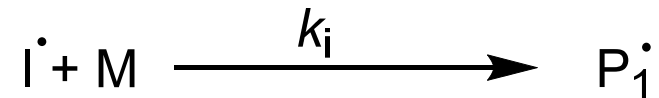
Chain(-Growth) Polymerization

Free-Radical Polymerization (FRP)

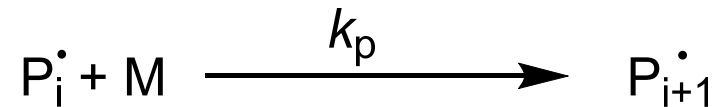
Initiator fragmentation



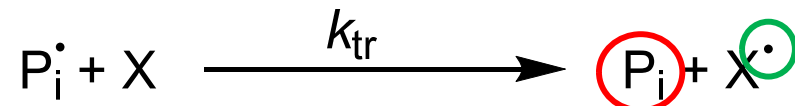
Initiation



Propagation



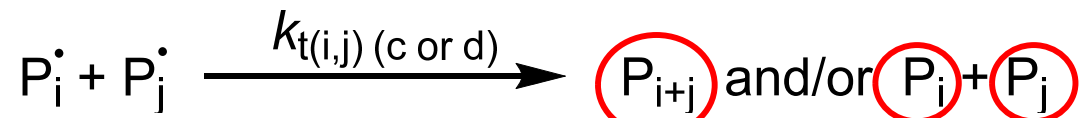
Transfer



Leads to dead chains

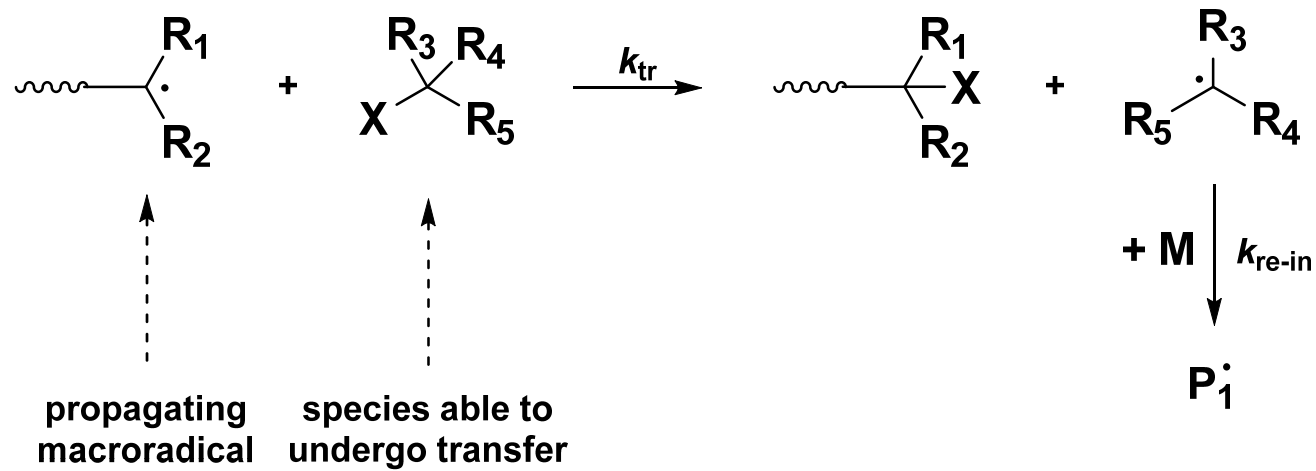
yet does not stop the polymerization!

Termination



Leads to dead chains

Transfer in Free-Radical Polymerization (FRP)



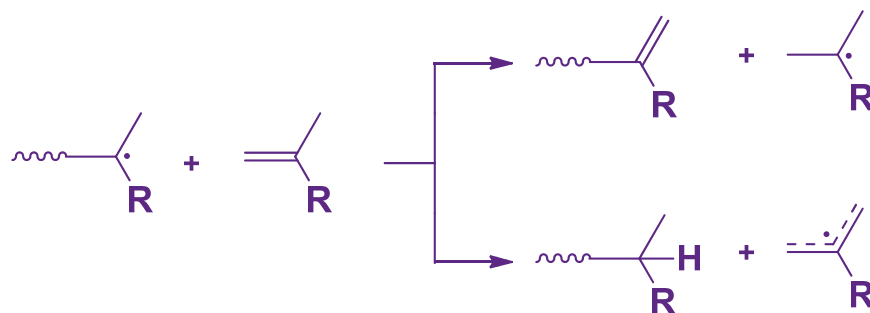
Transfer constant $C = \frac{k_{tr}}{k_p}$

Which species can undergo this type of reactions? Or "What can X be?"

Transfer in FRP

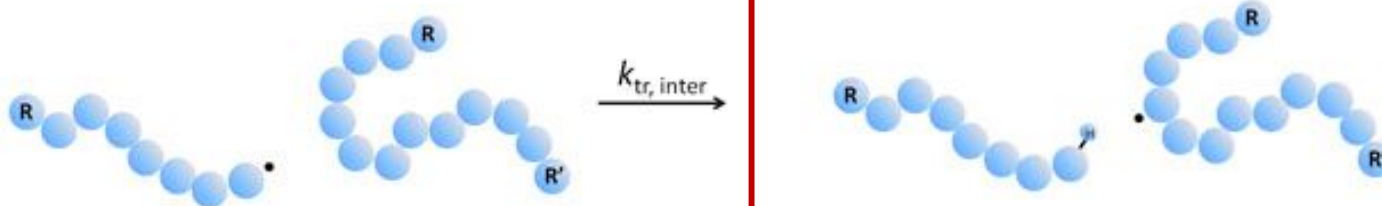
Several Types

To the monomer

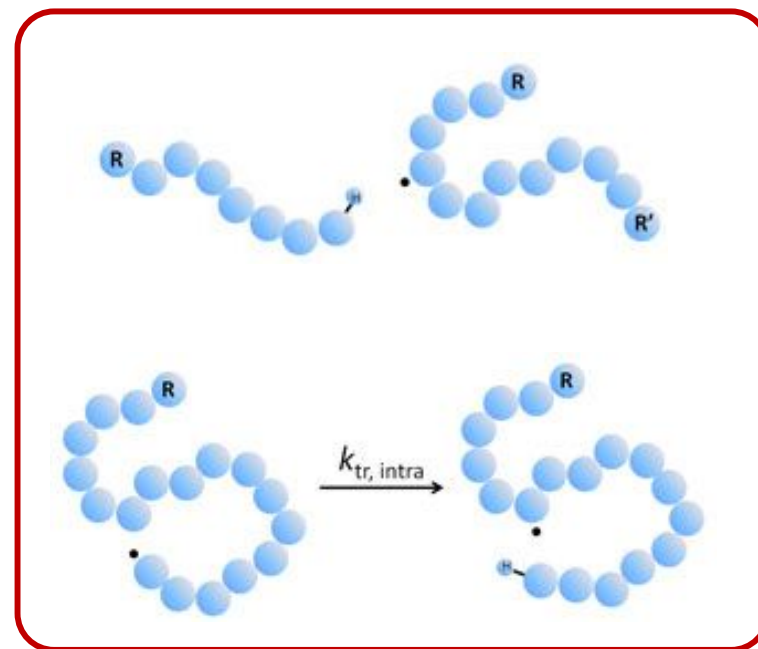
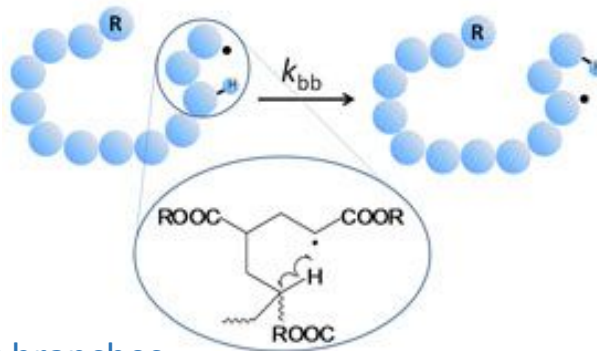


To the polymer

Intermolecular



Intramolecular



Random abstraction → long branches

Backbiting → short branches

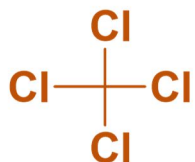
Transfer in FRP

Several Types

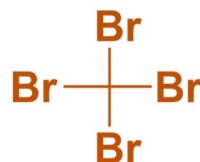
To a transfer agent

Sometimes, polymer chemists want to keep the molecular weight in a polymerization very low (to make so-called **oligomeric material**) and they deliberately add a transfer agent.

Typical transfer agents are:



Carbon tetrachloride



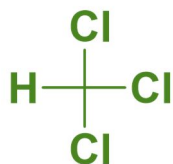
Carbon tetrabromide



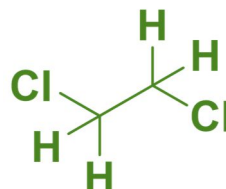
Dodecanethiol

Can you identify the group in the particular reagents that is actually transferred ?

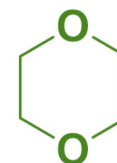
To the solvent



Chloroform



1,2-Dichloroethane



1,4-Dioxane

Retardation and Inhibition in FRP

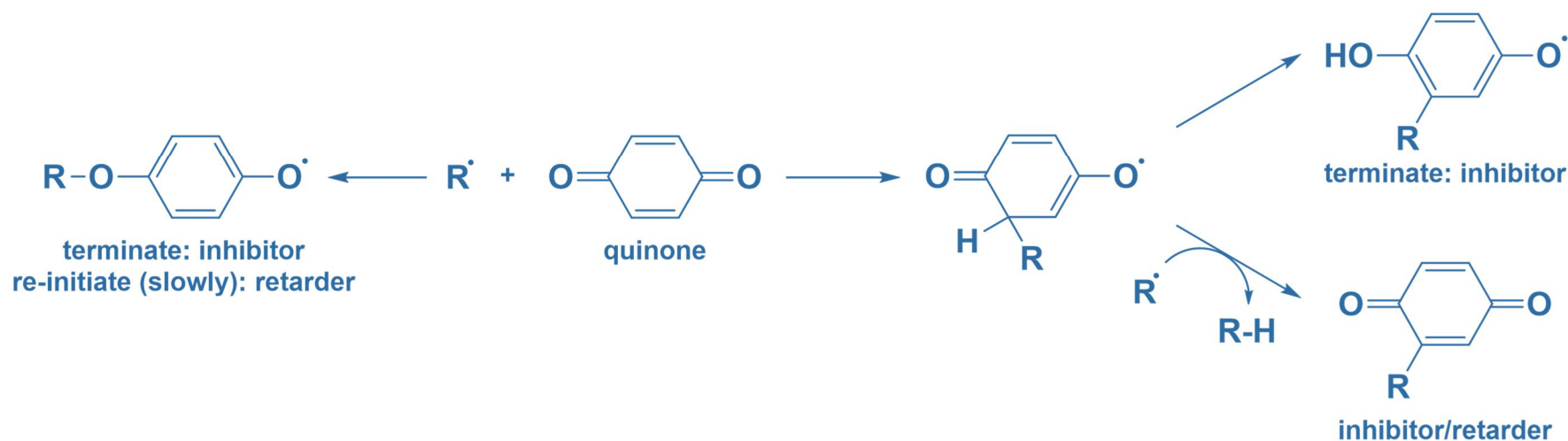
In the previous chain transfer section, we assumed that the small radical generated via the transfer process is able to re-initiate again at a reasonable rate.

However, some transfer agents yield radicals that will not re-initiate and thus slow down (**retardation**) or stop (**inhibition**) the polymerization.

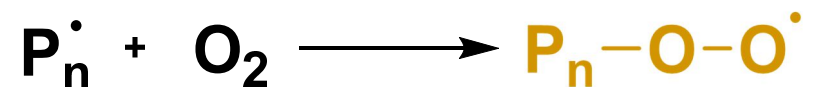
These types of chain transfer agents are called **retarders** and **inhibitors**, respectively.

Depending on the reaction conditions, an inhibitor may become a retarder and vice versa.

A good example of a retarder/inhibitor is quinone:

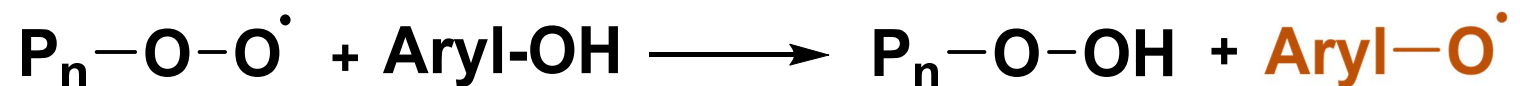


Another important retarder is oxygen, which forms **peroxy radicals** with the propagating radicals and are relatively unreactive in addition reactions.



The peroxy radicals mostly terminate with other radicals.

In addition, oxygen helps the inhibition action of phenols via the creation of the very unreactive **aryl-oxy radical**.



Thus, for a successful free radical polymerization, inhibitors and oxygen should be removed from the reaction mixture.

Inhibitors are not only bad though! ***They enable us to store monomers*** without them polymerizing when we don't want them to!

CHAPTER 1

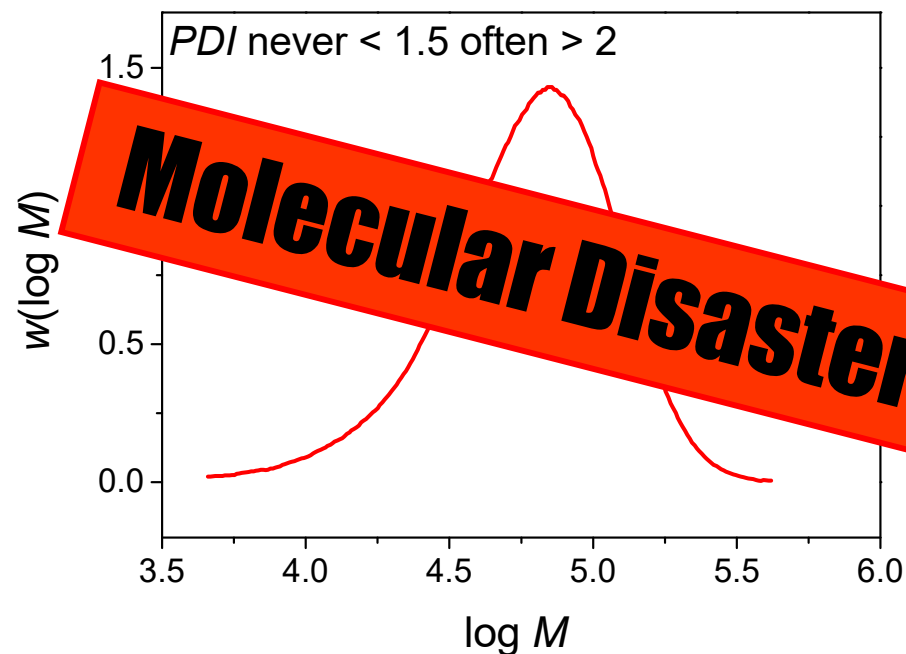
Reversible-deactivation radical polymerization (RDRP)

Prog. Polym. Sci. **2020**, 111, 101311

Polym. Chem. **2018**, 9, 4947

RDRP - The Concept

Free-Radical Polymerization (FRP)



Broad Molecular Weight Distribution

Poor Functionality

What Can We Do ?

However, FRP bears many great advantages! It is:

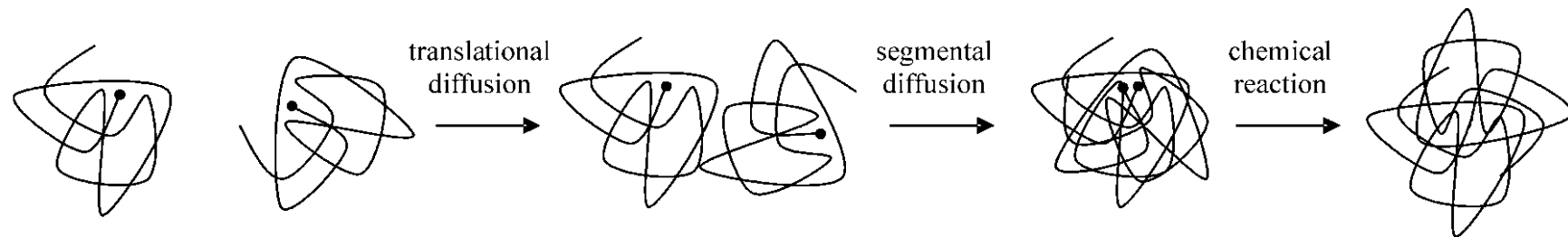
- relatively insensitive to environment (solvent) polarity change;
- tolerant to a large variety of functional groups;
- applicable to water-borne systems;
- a versatile technique, which can be triggered through various processes.

Heat, light, redox, gamma rays, plasma, microwave, ultrasound

RDRP - The Concept

Free-Radical Polymerization (FRP)

Main issue: Radicals do terminate...



The formed polymers feature broad distributions and are dead, having no synthetic handle.

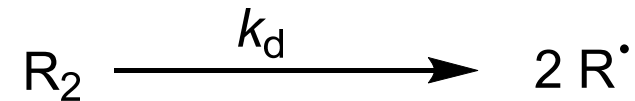


We need to extend the lifetime of each propagating radical.

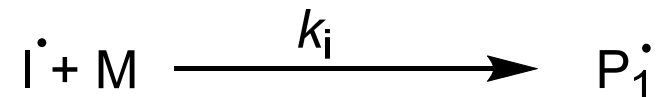
RDRP - The Concept

Free-Radical Polymerization (FRP)

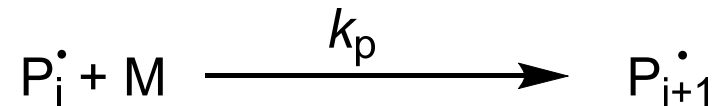
Initiator fragmentation



Initiation



Propagation



Transfer

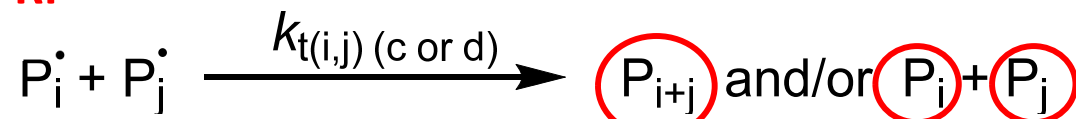


Leads to dead chains

yet does not stop the polymerization!

main chain breaking-reaction-in-RP

Termination



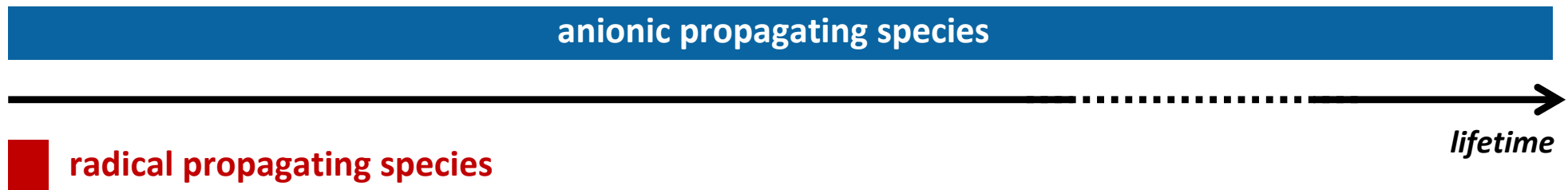
Leads to dead chains

RDRP - The Concept

Free-Radical Polymerization (FRP)

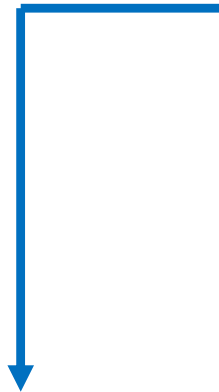
A **radical** lives for *less than a second*, at best a few seconds...

... in contrast, propagating species in **anionic** polymerization can be kept as such for **hours, days, weeks, months...**



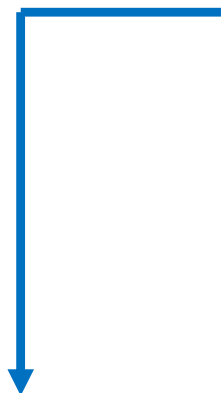
RDRP - The Concept

Limiting irreversible termination events → Taming radicals



RDRP - The Concept

Limiting irreversible termination events → Taming radicals



Reversible Termination

Nitroxide-Mediated
Polymerization (NMP)

Atom-Transfer Radical
Polymerization (ATRP)



Reversible Transfer

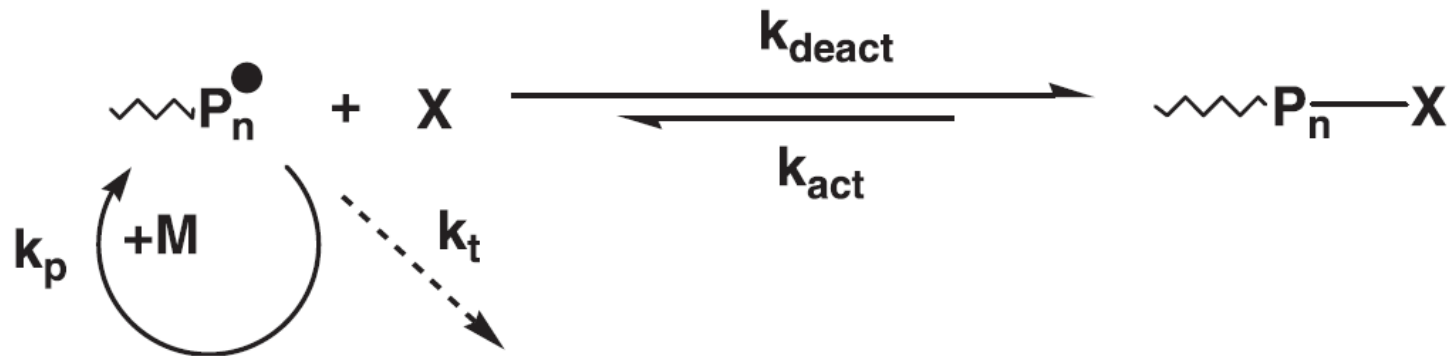
Iodine-Transfer Polymerization

Reversible Addition–Fragmentation
Chain-Transfer (RAFT)
Polymerization



RDRP - The Concept

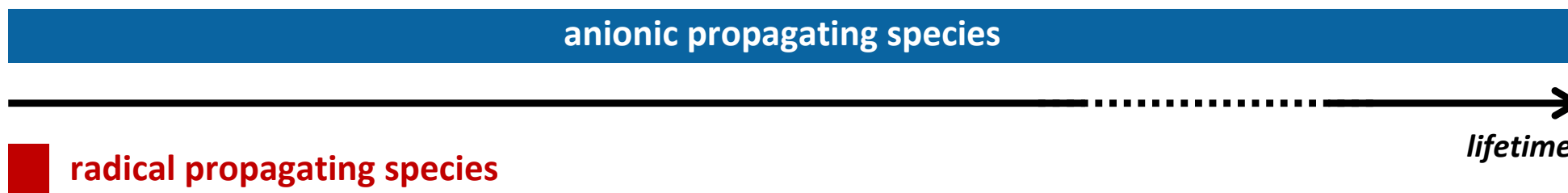
Limiting irreversible termination events → Taming radicals



RDRP - The Concept

Limiting irreversible termination events → Taming radicals

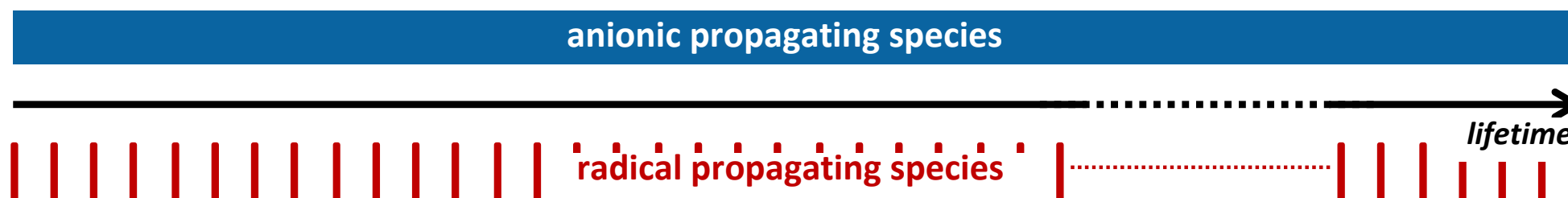
Idea: Chopping up the active lifetime of a radical to artificially increase the overall lifetime



RDRP - The Concept

Limiting irreversible termination events → Taming radicals

Idea: Chopping up the active lifetime of a radical to artificially increase the overall lifetime



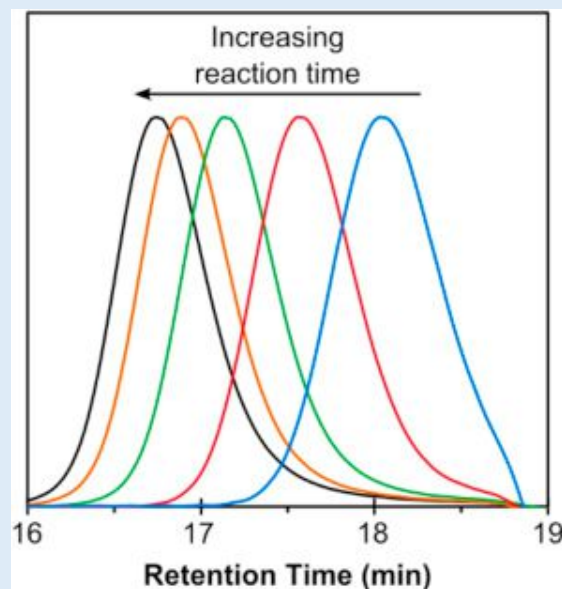
RDRP - The Concept

Historical Perspective

- 1956** *M. Szwarc*
Conception of 'Living' Polymerization, Anionic Polymerization
- 1968** *E. Borsig*
First Example of Tetraarylethane-Mediated Living Free Radical Polymerization
- 1976** *K. Matyjaszewski*
Ring-Opening Cationic Living Polymerization, Active/Dormant Equilibrium
- 1979** *Daikin Industries Ltd., Japan*
Patent on Iodine-Mediated Radical Polymerization
- 1982** *T. Otsu*
Iniferter Polymerization
- 1985** *D. Solomon, E. Rizzardo*
Nitroxide-Mediated Polymerization (NMP)
- 1995** *M. Sawamoto, K. Matyjaszewski*
Atom-Transfer Radical Polymerization (ATRP)
- 1998** *E. Rizzardo, G. Moad*
Reversible Addition-Fragmentation Chain-Transfer (RAFT) Polymerization

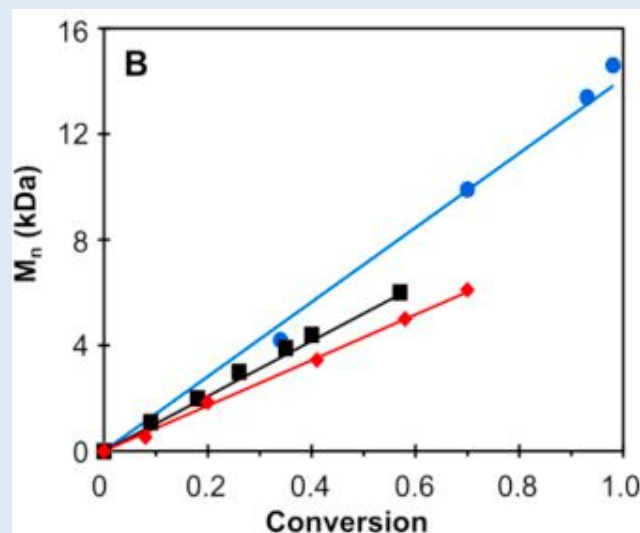
RDRP - The Concept

Criteria to assess whether a polymerization is controlled/living

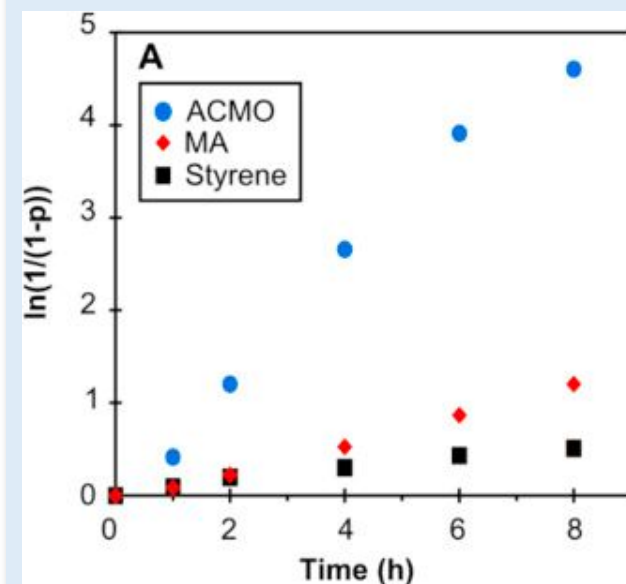


Molar masses increase with time/conversion

Low dispersity (typically, $\bar{D} < 1.2$)



Linear growth with conversion



Constant propagating species concentration

Fast initiation/transfer

Chains are dormant $\rightarrow$ Quantitative chain extension (living!) $\rightarrow$ end-group transformation + block copolymers

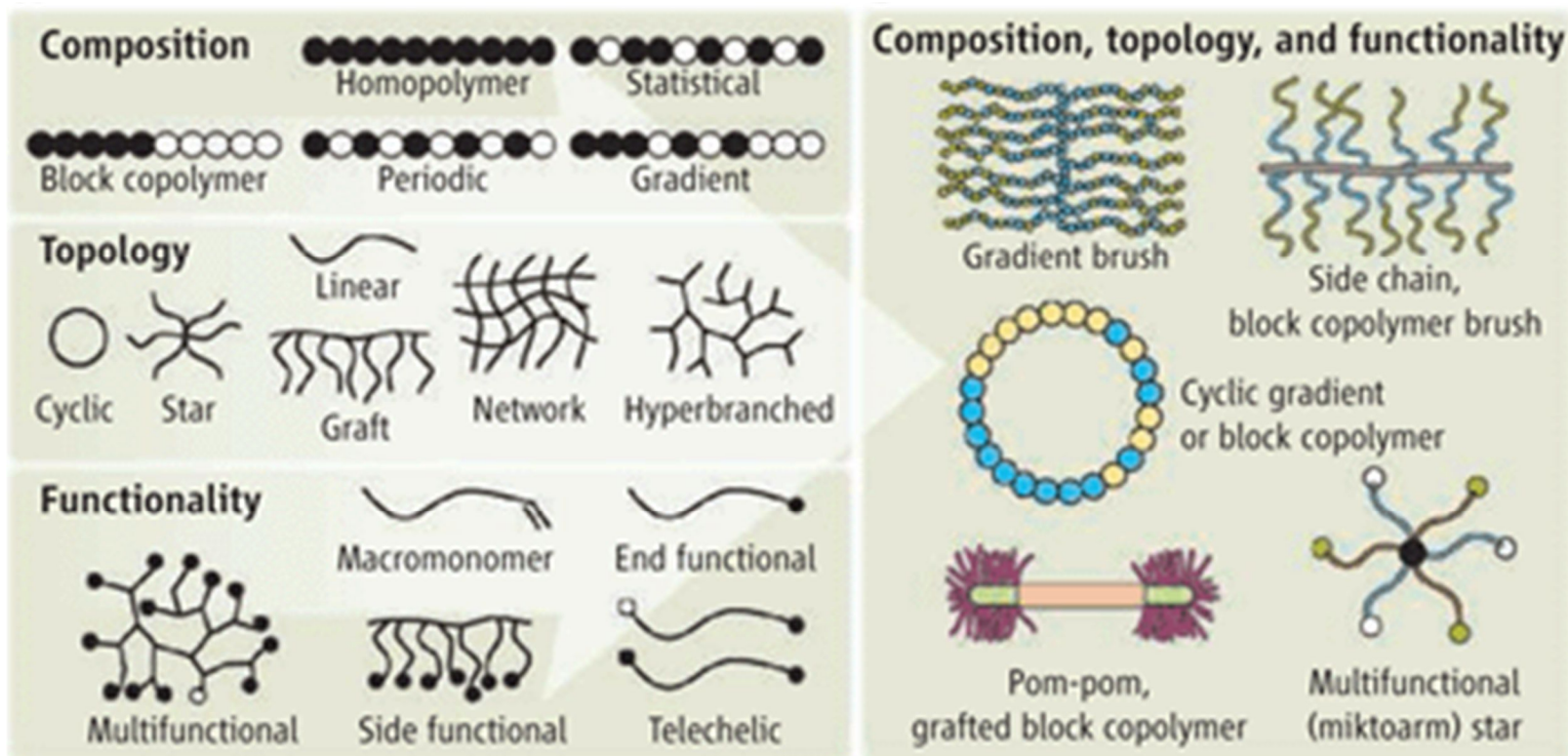
RDRP - The Revolution

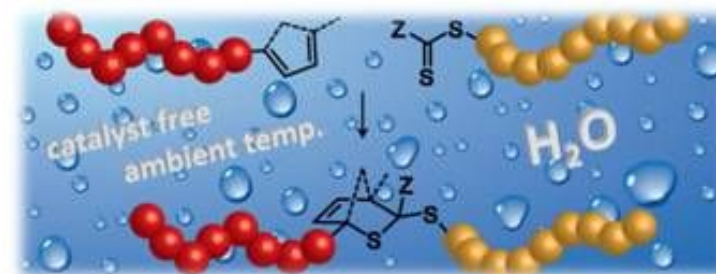
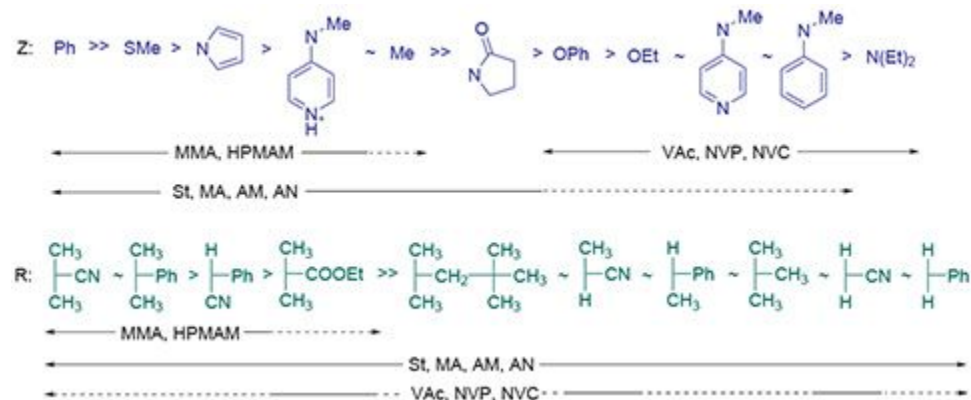
What RDRP has enabled (further, after anionic polymerization)

- Homogeneous polymer chain properties: example of copolymerization
- Block copolymers!
- Architectures!
- Hybrids!

RDRP - The Revolution

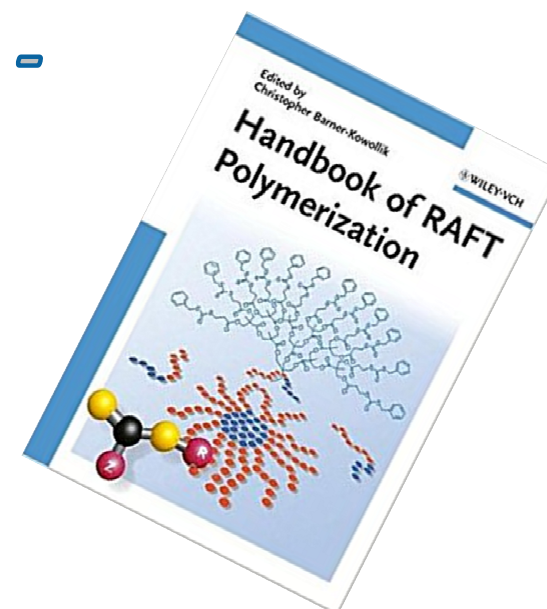
What RDRP has enabled (further, after anionic polymerization)





RDRP

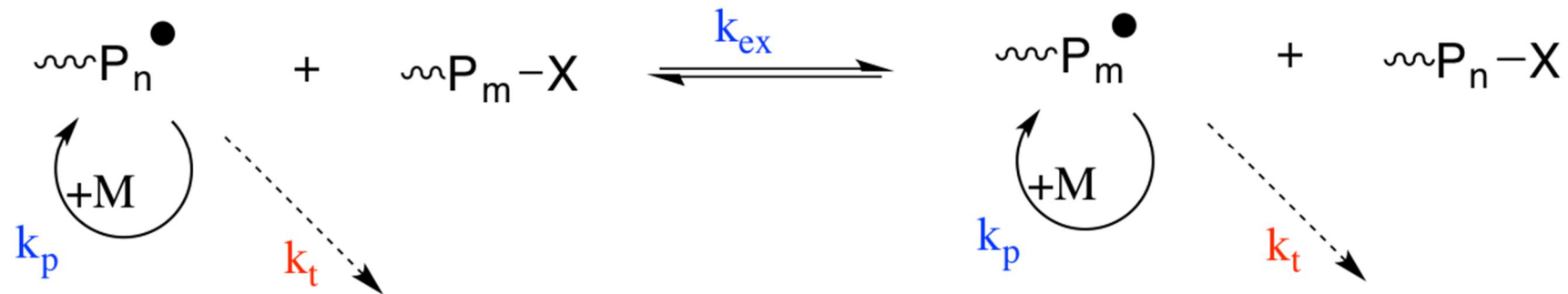
- Reversible Addition-Fragmentation Transfer Polymerization -



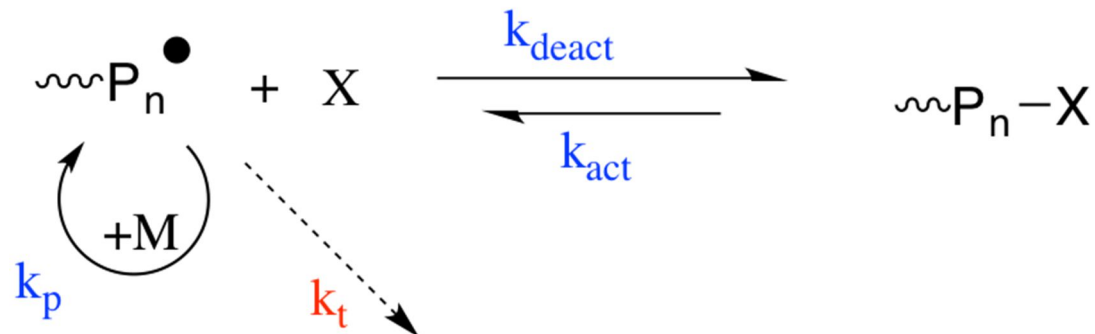
Macromolecules **2017**, 50, 7433

A semi-symmetrical equilibrium

Degenerative chain transfer mechanism



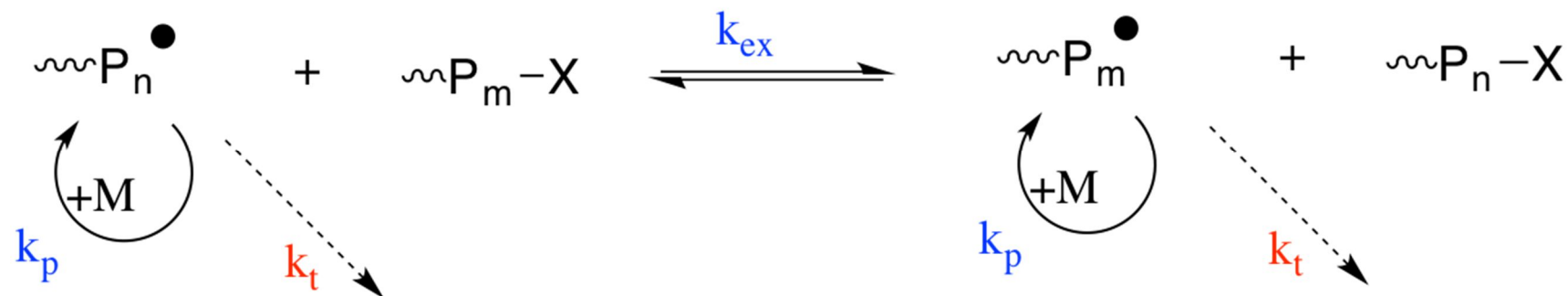
$\neq$ reversible termination mechanism



Reversible Addition-Fragmentation Transfer (RAFT) Polym.

A semi-symmetrical equilibrium

Degenerative chain transfer mechanism

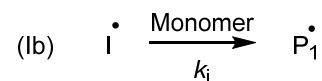
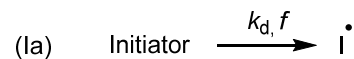


$X = -I \quad \Longrightarrow \quad \text{Iodine-Transfer Polymerization}$

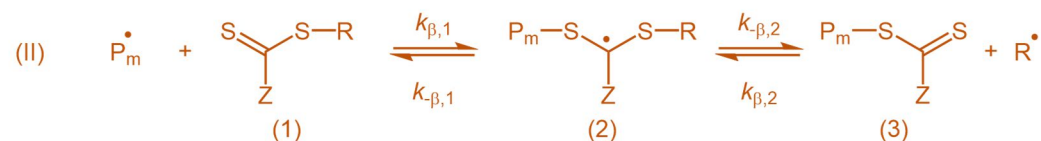
$\begin{array}{c} -S \\ \diagup \\ C=Z \\ \diagdown \\ S \end{array} \quad \Longrightarrow \quad \text{RAFT Polymerization}$

Detailed mechanism

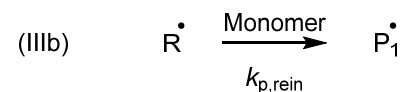
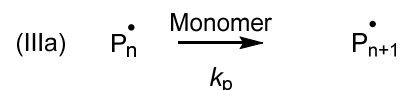
I. INITIATION



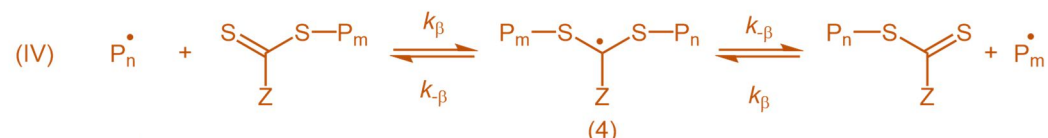
II. PRE-EQUILIBRIUM



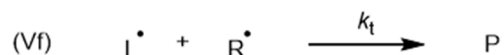
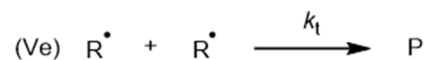
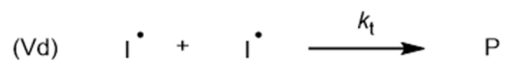
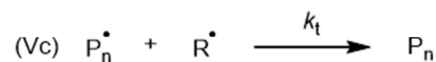
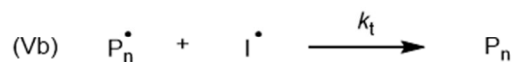
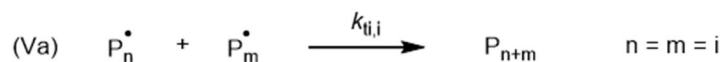
III. PROPAGATION



IV. CORE EQUILIBRIUM



V. TERMINATION



Conventional free-radical polymerization plus two superimposed equilibria:

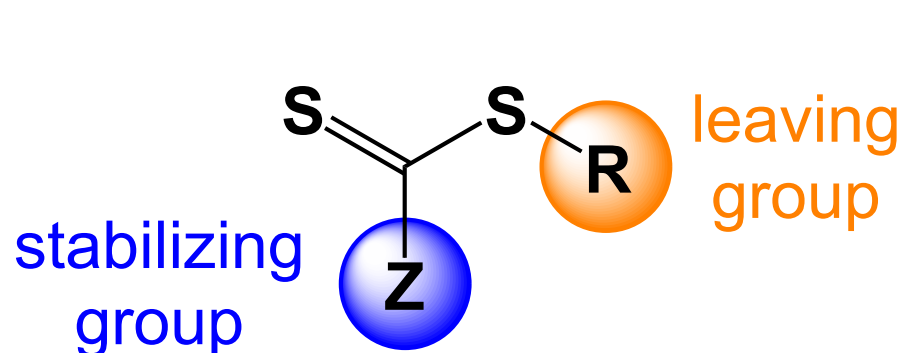
- **Pre-equilibrium:** Transformation of the initial RAFT agent into polymeric RAFT Agent
- **Main-equilibrium:** Chain equilibration effecting living characteristics

In an **ideal RAFT process** the rate of polymerization, R_p ,
is identical to that of the conventional process under identical reaction conditions

However, a range of RAFT-mediated polymerizations display inhibition and retardation phenomena.

Reversible Addition-Fragmentation Transfer (RAFT) Polym.

Typical RAFT agents and their structural variety



	Z
Dithioesters	{ Ph CH ₃
Dithiocarbamates	NR ₂
Xanthates	OR
Trithiocarbonates	SR

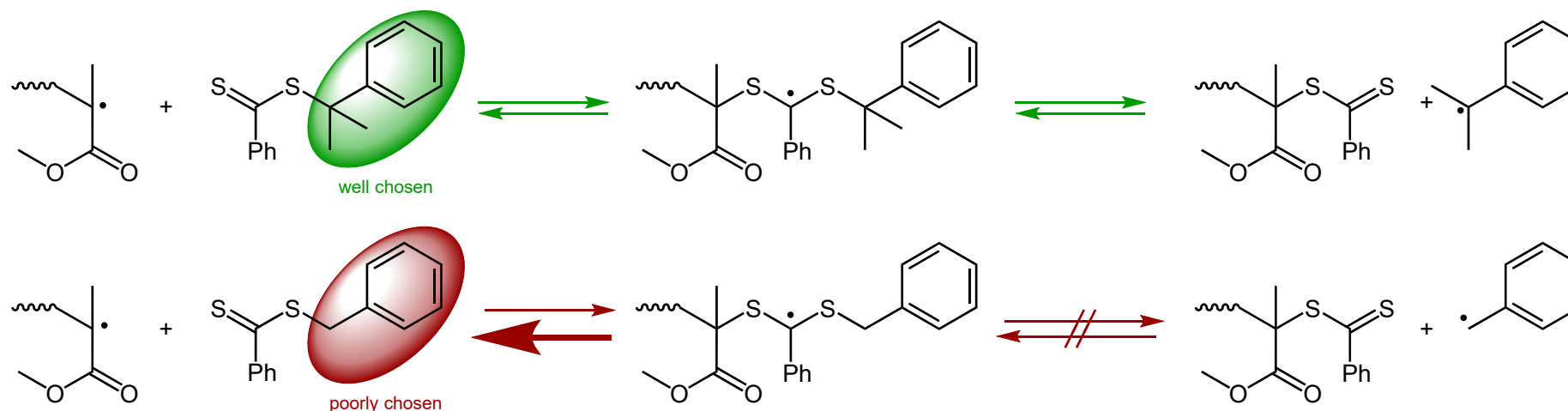
Z modifies addition and fragmentation rates.

R is a good leaving group AND must be able to reinitiate a chain growth.

R
CH₂ Ph
CH₂CN
CH(CH₃)COOH
C(CH₃)(CN)CH₂CH₂COOH
C(CH₃)₂Ph
C(CH₃)₂CN

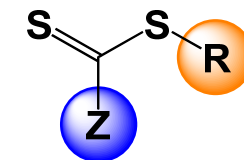
Typical RAFT agents and their structural variety

The RAFT agent and monomer have to be carefully selected for an efficiently functioning RAFT process:



Reversible Addition-Fragmentation Transfer (RAFT) Polym.

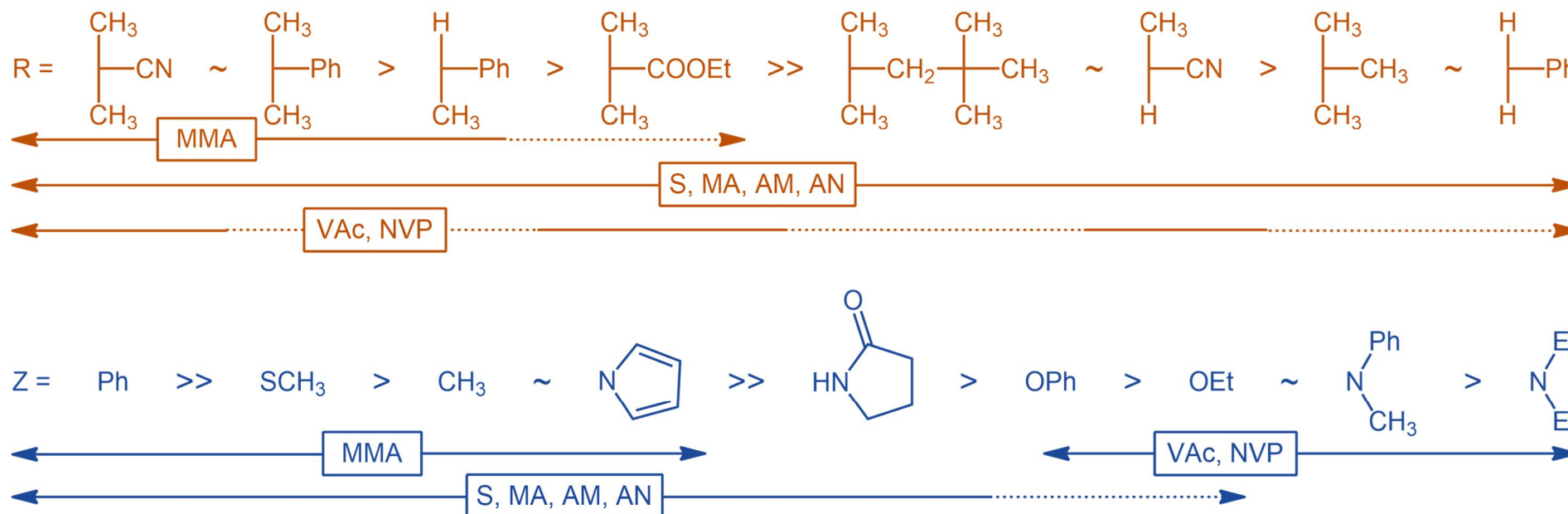
Typical RAFT agents and their structural variety



Typical RAFT reaction conditions are **identical** to those of the associated **conventional FRP**.

Nearly all monomers can be polymerized in a living fashion via the RAFT process – provided the RAFT agent is judiciously chosen.

Purification of the resulting polymers is **simple** – they are however colored (red, pink, yellow).

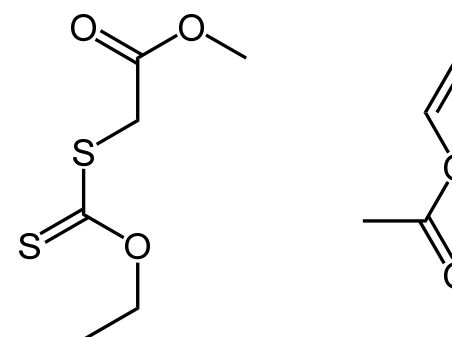
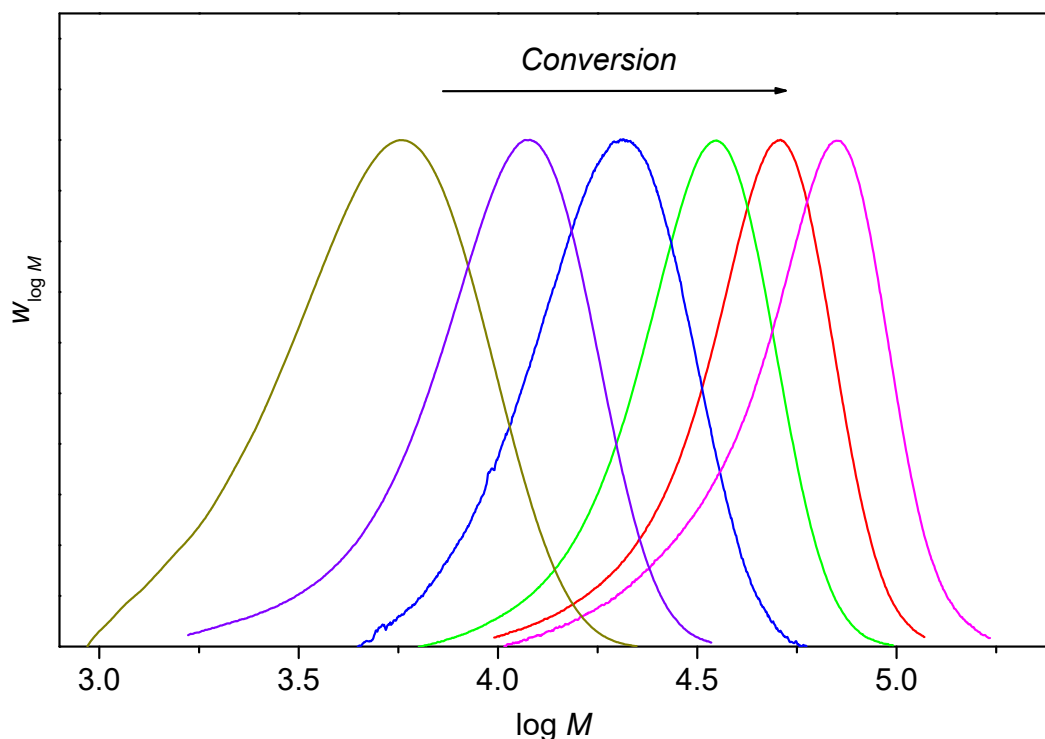


The (RAFT/)MADIX process

MADIX = **M**acromolecular **d**esign by **i**nterchange of **x**anthates

Essential for the polymerization of vinyl esters and vinyl amides/amines (also vinylphosphonic acid)

Typical example of LFRP of vinyl acetate via RAFT/MADIX



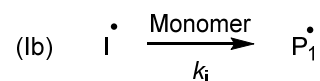
60 °C, AIBN

Other important monomers requiring xanthates: *N*-vinylpyrrolidone, *N*-vinylcaprolactam, *N*-vinylcarbazole, vinylidene fluoride

Initiating RAFT Polymerization

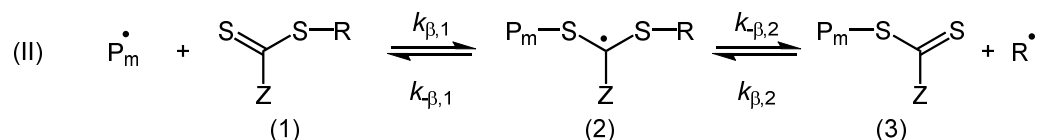
How do I prepare my polymerization mixture?

I. INITIATION



- Thermal initiator (styrene - autoinitiation)

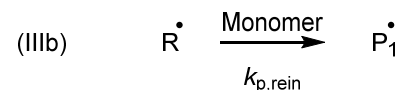
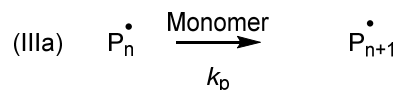
II. PRE-EQUILIBRIUM



- Redox (ATRP initiator could work!)

- Photoinitiator

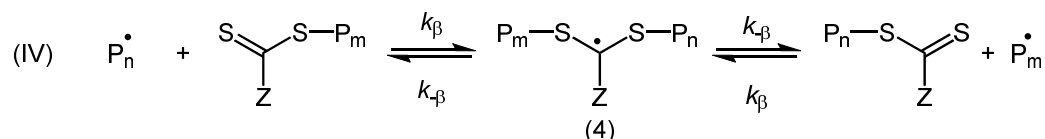
III. PROPAGATION



- Microwave

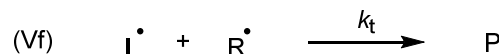
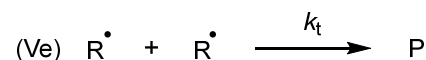
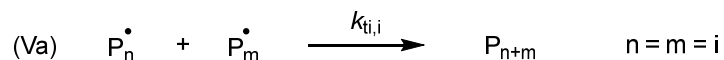
- Gamma rays

IV. CORE EQUILIBRIUM



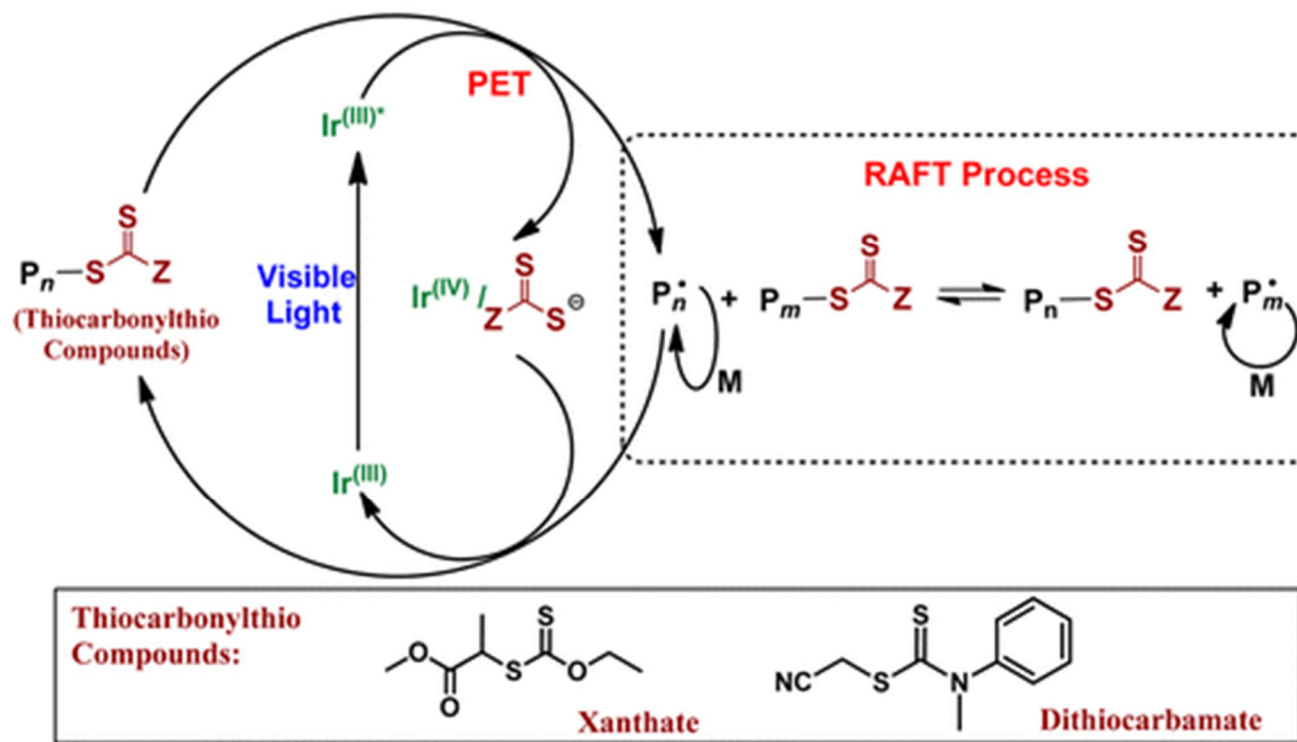
- Plasma

V. TERMINATION



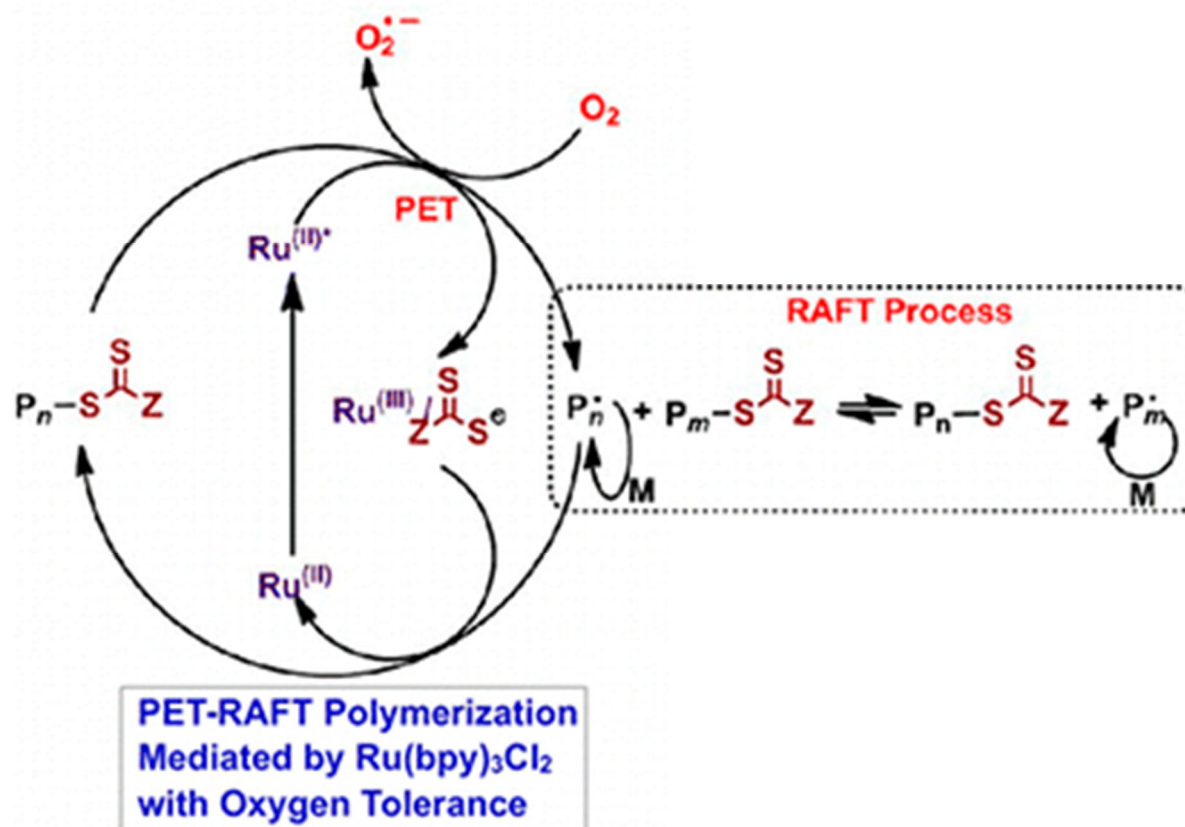
Initiating RAFT Polymerization

Photoelectron/energy transfer (PET) RAFT

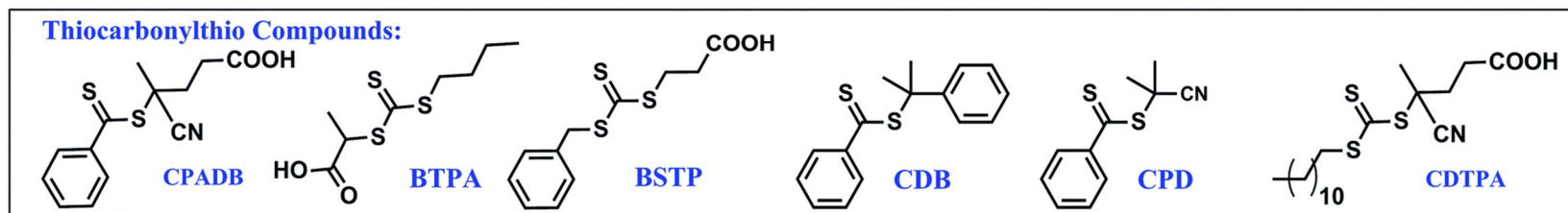
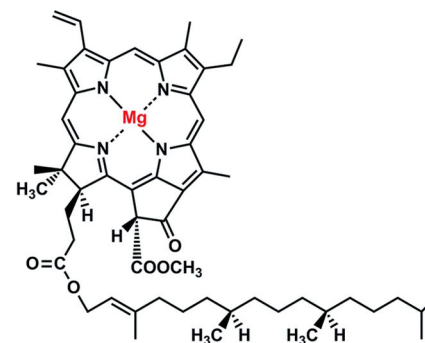
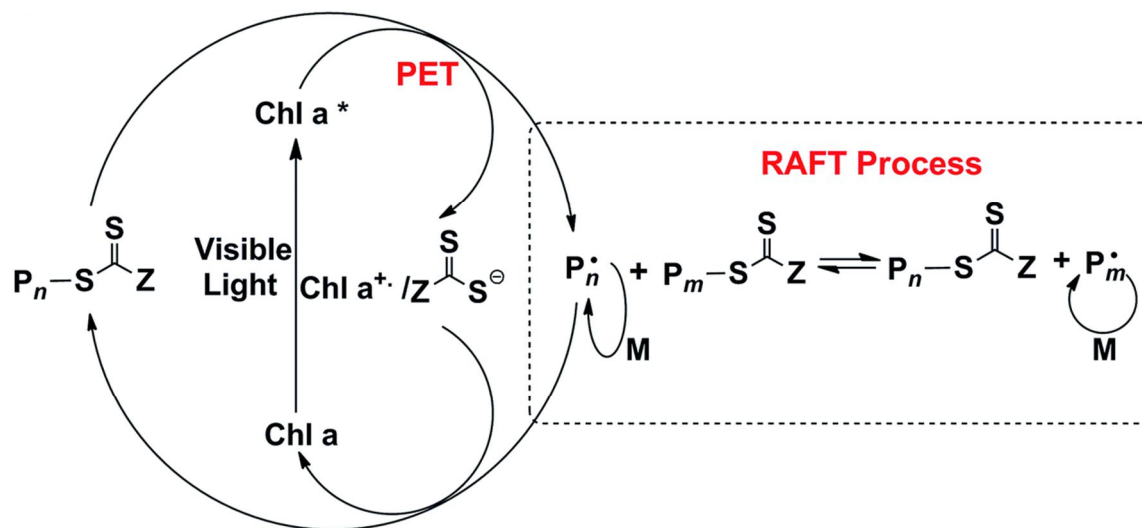


Initiating RAFT Polymerization

Photoelectron/energy transfer (PET) RAFT



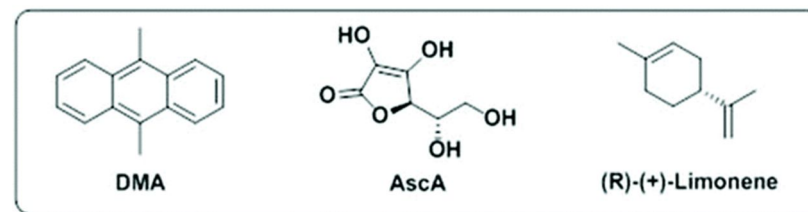
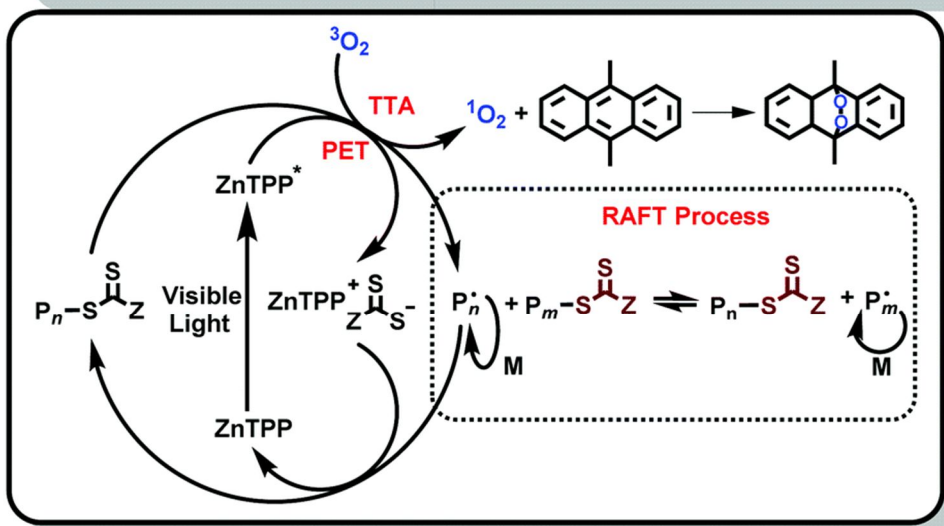
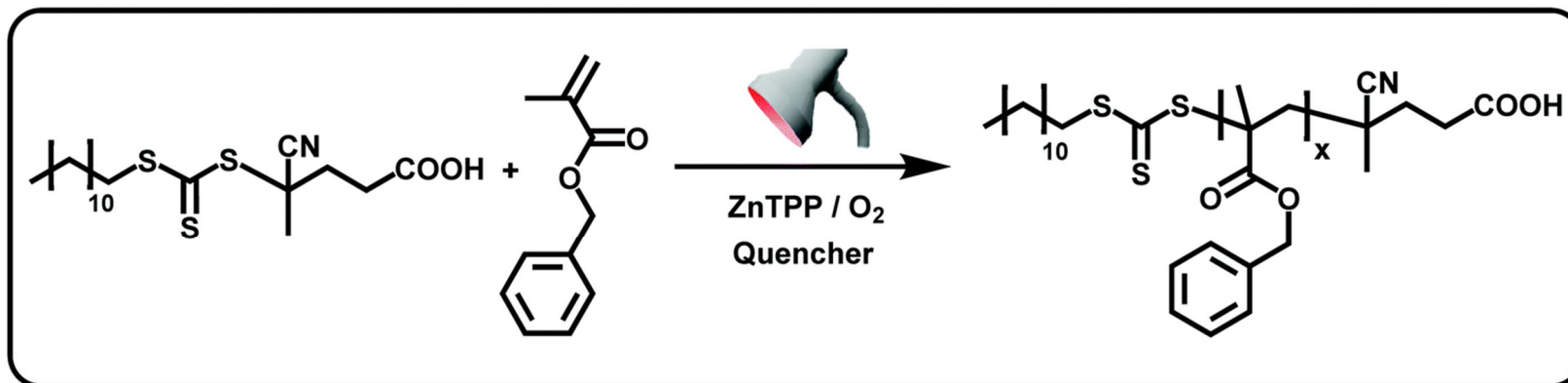
Polymerization performed in DMSO, which is a good singlet oxygen quencher



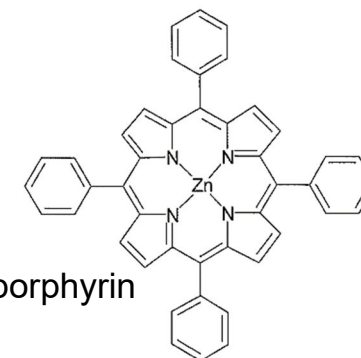
Much cheaper than the previous metal complexes... Unfortunately, PET-RAFT sensitive to oxygen.

Initiating RAFT Polymerization

Photoelectron/energy transfer (PET) RAFT



ZnTPP
Zinc tetraphorphyrin

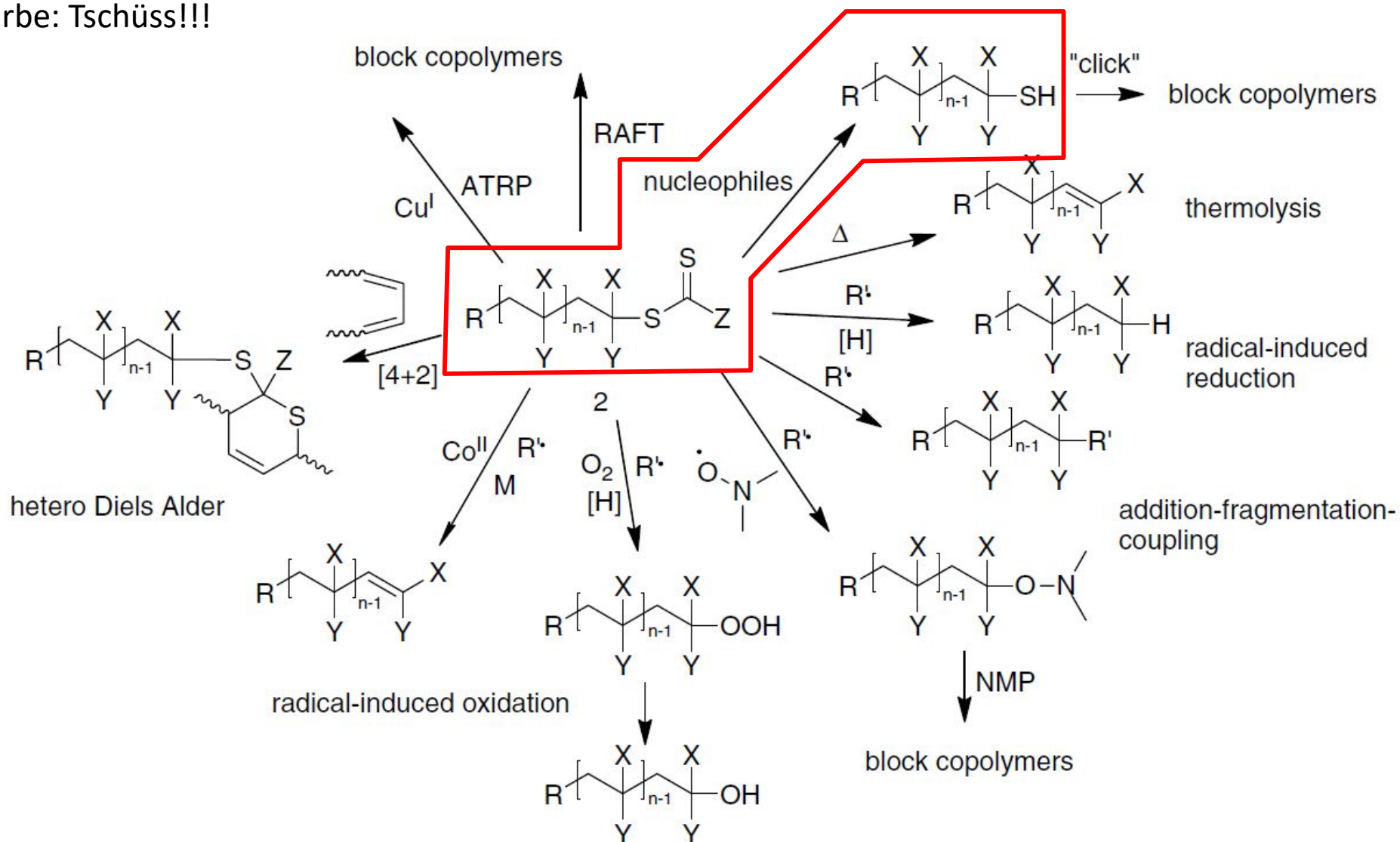


Transfer to other solvents... addition of singlet oxygen quenchers

End-Group Transformation of RAFT Polymers

Reactions at the thiocarbonylthio group

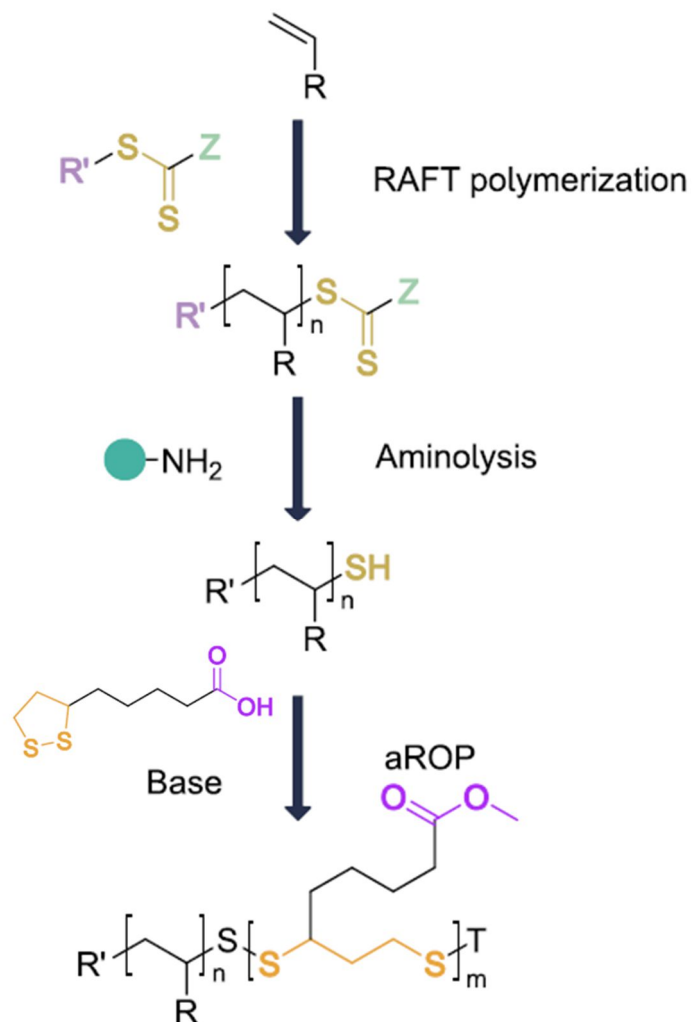
Farbe: Tschüss!!!



(R'· = radical; [H] = hydrogen donor; M = monomer)

End-Group Transformation of RAFT Polymers

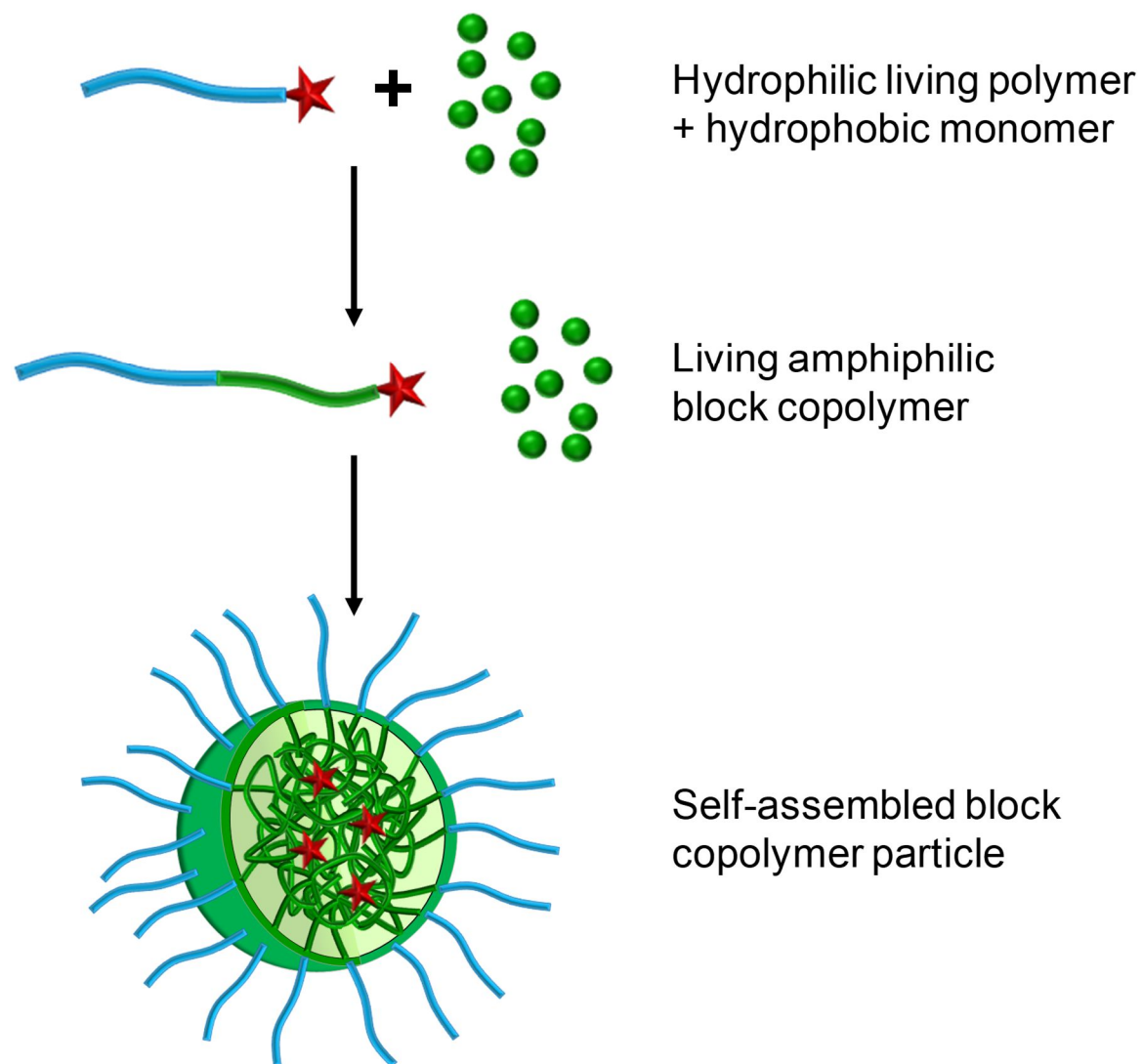
Reactions at the thiocarbonylthio group



Adapted from the MSc thesis of S. Hylla (AK Delaittre, HHU, 2025)

Polymerization-Induced Self-Assembly

Emulsion/dispersion polymerization with RAFT polymers as stabilizers



Polymerization-Induced Self-Assembly

Emulsion/dispersion polymerization with RAFT polymers as stabilizers

